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Blood 144 (2024) 7409–7410

The 66th ASH Annual Meeting Abstracts

ONLINE PUBLICATION ONLY

723.ALLOGENEIC TRANSPLANTATION: LONG-TERM FOLLOW-UP, COMPLICATIONS, AND DISEASE RECURRENCE

Impact of Systemic Antibiotics Use on Outcomes after Allogeneic Hematopoietic Stem Cell Transplantation

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Introduction: Post-transplant complications lead to significant morbidity and mortality in the management of patients undergoing allogeneic hematopoietic stem cell transplant (allo-HCT). Broad-spectrum systemic antimicrobials and better prophylaxis strategies have reduced infection-related morbidity and mortality in allo-HCT patients. However, broad-spectrum antibiotic use disrupts the microbiome and immunome, which may result in poor post-transplant outcomes. We aimed to evaluate the impact of systemic antibiotics on allo-HCT outcomes.

Methods: A retrospective multicenter analysis was conducted, including allo-HCT patients in the publicly available Center for International Blood and Marrow Transplant (CIBMTR) registry from 2013 to 2018 using P5646 data by Ramathan et al. Chi-square and t-tests were used to compare categorical and continuous baseline demographics. Multivariate Cox regression analyses were used to calculate hazard ratios (HR) with 95% confidence intervals (CI) for overall survival (OS), disease-free survival (DFS), relapse, non-relapse mortality (NRM), acute graft-versus-host disease (aGVHD) and chronic GVHD (cGVHD), engraftment, and infection-related mortality between the two groups. Statistical analysis was conducted using R version 4.16, with statistical significance defined as p <0.05.

Results: Among 7524 allo-HCT patients, 5779 were administered broad-spectrum systemic antibiotics (SA) while 1745 were not given systemic antibiotics (NSA). The median age was 58.58 years (range: 2.03-82.67) in the SA group and 56.18 years (range: 2.00 - 77.86) in the NSA group, and 58% were men in both groups. Ethnicities were Caucasian (76%), Hispanic (8%), African American (6%), and others (10%). The primary disorders included acute myeloid leukemia (47%), myelodysplastic syndrome (35%), and acute lymphoblastic leukemia (18%). Myeloablative conditioning was used in 53% of patients. Donor types were matched unrelated (49.5%), matched related (34%), and cord blood (16%). Graft sources were bone marrow (15%), peripheral blood stem cells (68%), and cord blood (16%). GVHD prophylaxis regimens included Tacrolimus and Cyclosporine-based (77%), post-transplant Cyclophosphamide-based (9%), Ex vivo T-cell depletion or CD34 selection (3%), and others (11%). In the multivariate regression analyses, systemic antibiotics independently predicted poor OS (55% SA vs 57% NSA; HR 1.13, 95% CI 1.01-1.22, p=0.026). Higher incidence of NRM (19% SA vs 15% NSA; HR 1.32, 95% CI 1.12-1.55, p=0.001), grade 2-4 aGVHD (41% SA vs 34% NSA; HR 1.24, 95% CI 1.12-1.38, p<0.001), and cGVHD (40% SA vs 39% NSA; HR 1.12, 95% CI 1.01-1.24, p=0.040) were observed with SA use. The two groups had no significant difference in DFS, relapse, infection-related mortality, and engraftment rates. The outcomes analyses were adjusted for the significant predictors identified in the univariate analyses.

Conclusion: Systemic antibiotics use in allo-HCT patients is associated with higher morbidity and mortality. Judicious use and de-escalation of systemic antibiotics when safe may help to improve post-transplant outcomes. Future research exploring microbiome restoration could result in better post-transplant outcomes.

Disclosures Hamadani: *Byondis:* Consultancy; *Omeros:* Consultancy; *Myeloid Therapeutics:* Speakers Bureau; *Forte Biosciences:* Consultancy; *Sanofi Genzyme:* Speakers Bureau; *AstraZeneca:* Speakers Bureau; *Autolus:* Consultancy; *BeiGene:*

Speakers Bureau; *DMC, Inc:* Speakers Bureau; *CRISPR:* Speakers Bureau; *BMS:* Consultancy; *Spectrum Pharmaceuticals:* Research Funding; *ADC Therapeutics:* Consultancy, Research Funding, Speakers Bureau; *CRISPR:* Consultancy; *Genmab:* Consultancy; *Astellas Pharma:* Research Funding; *Takeda:* Research Funding; *Allovir:* Consultancy; *Caribou:* Consultancy; *Genentech:* Speakers Bureau; *Kite Pharma:* Consultancy, Speakers Bureau; *AbbVie:* Consultancy. **McGuirk:** *Autolus:* Consultancy; *Kite:* Consultancy; *Envision:* Consultancy; *Allo Vir:* Consultancy; *CRISPR therapeutics:* Consultancy; *Sana technologies:* Consultancy; *Caribou bio:* Consultancy; *BMS:* Consultancy; *Novartis:* Consultancy; *NEKTAR therapeutics:* Consultancy; *Legend biotech:* Consultancy. **Mushtaq:** *Iovance Biotherapeutics:* Research Funding.

https://doi.org/10.1182/blood-2024-193482